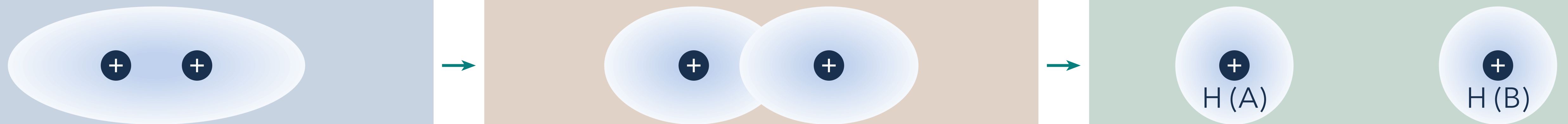
Bond breaking is central to chemical reactions; H_2 is the simplest test of whether DFT gets the electron rearrangement right.

1 Stretch the bond: a second configuration becomes essential



BONDED

Mostly two electrons in the bonding orbital

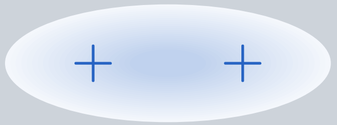
STRETCHED

Antibonding-pair configuration gains weight

SEPARATED

One electron on each neutral H atom

σ_g: bonding

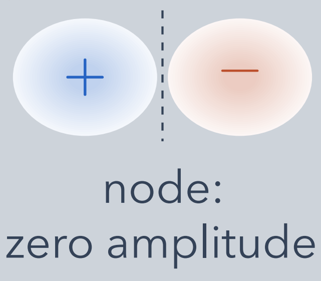


g = gerade (even)

Invert through midpoint: orbital keeps its sign.

1s waves add → density builds between nuclei.

σ_u: antibonding (also σ_u^*)



u = ungerade (odd)

Invert through midpoint: orbital flips its sign.

1s waves subtract → a node between nuclei.

+/− show wavefunction sign (phase), not electric charge. Inversion maps each point through the bond center.

READ THE SYMBOLS

σ: unchanged by rotation around the H–H axis.

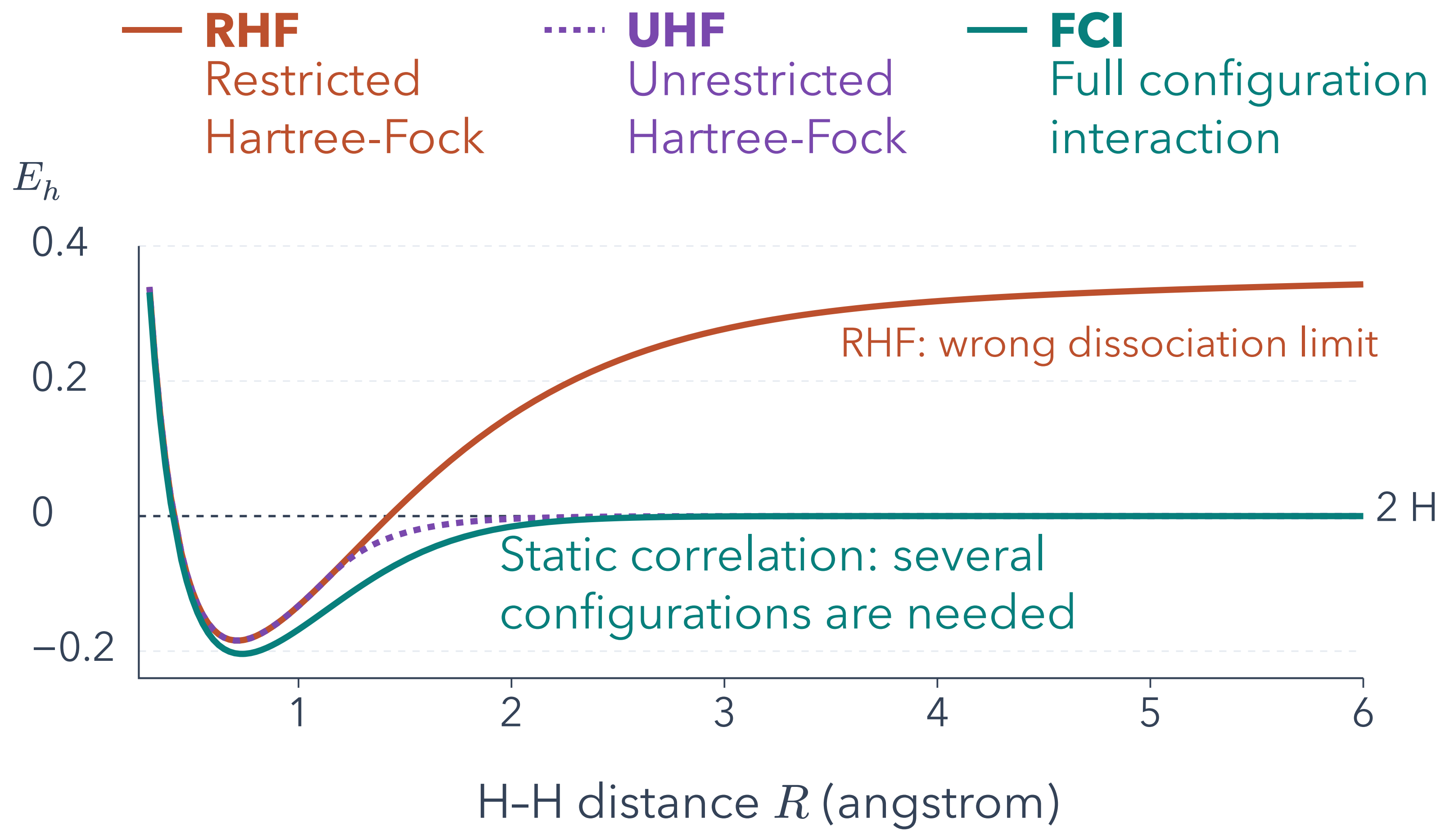
σ_g^2: two opposite-spin electrons in σ_g.

Configuration = an orbital-occupation pattern.

g/u label symmetry, not bonding in general; an asterisk marks antibonding.

2 Watch the energy and the real-electron occupations

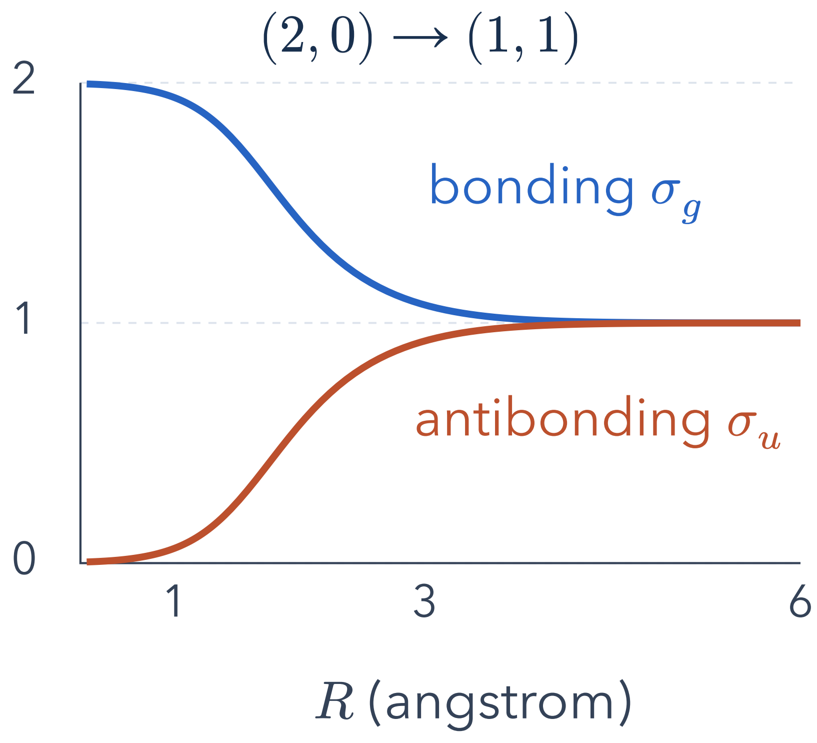
Energy relative to two separated H atoms



Natural occupations

Electrons per orbital; total = 2.

Bonding/antibonding: $(2, 0) \longrightarrow (1, 1)$



Basis = allowed orbital building blocks; STO-3G uses one 1s (lowest atomic orbital) per H. FCI mixes all allowed configurations and is **exact in this basis**.

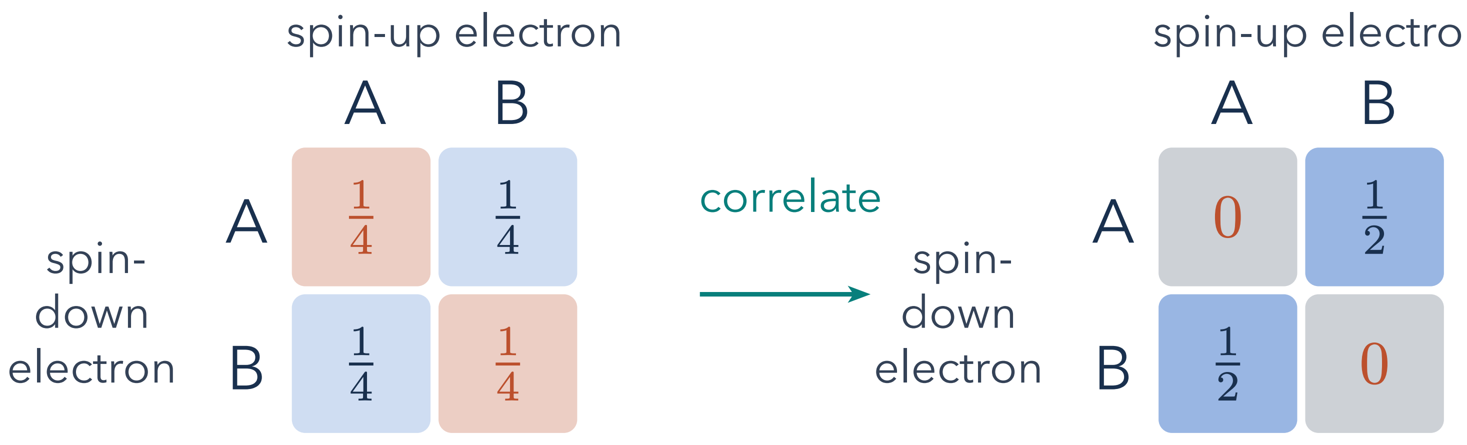
3 Separated atoms: density can hide wrong pair probabilities

Restricted σ_g^2 reference

Restricted: both spins share one spatial orbital.

Correlated singlet

Singlet: spins combine to zero. Measured along any shared axis, they give opposite results.



spin-down electron \ spin-up electron	A	B
A	$\frac{1}{4}$	$\frac{1}{4}$
B	$\frac{1}{4}$	$\frac{1}{4}$

spin-down electron \ spin-up electron	A	B
A	0	$\frac{1}{2}$
B	$\frac{1}{2}$	0

50% ionic weight

0% ionic weight

Ionic weight = chance both electrons are on the same atom. Orange diagonal: charged $H^- + H^+$; blue off-diagonal: neutral $H + H$.

Cancel the ionic amplitudes

A/B: 1s orbitals on atoms A/B; (1)/(2): electron labels.

$$\sigma_g = \frac{A+B}{\sqrt{2}} \qquad \sigma_u = \frac{A-B}{\sqrt{2}}$$

$$\Psi_S = \tfrac{1}{\sqrt{2}}\left(\sigma_g(1)\sigma_g(2) - \sigma_u(1)\sigma_u(2)\right)$$

$$= \tfrac{1}{\sqrt{2}}\left(A(1)B(2) + B(1)A(2)\right)$$

Ψ_S: the pair's spatial wave; $|\Psi_S|^2$: probability density.

Superposition: add waves first, then square to get probabilities. Same-atom terms cancel; only one-on-each-atom terms survive.

4 What this means for density-functional theory (DFT)

KOHN-SHAM ORBITAL

$$\varphi_{\mathrm{KS}} = \sqrt{\tfrac{n}{2}}$$

n: electron density · φ_{KS}: model orbital

Its occupation stays 2. Exact XC supplies correlation; this model electron pair is not the real state.

UHF: MIXED SPIN STATE

UHF lets up/down spins use different orbitals.

It reaches two neutral H atoms, but the spins no longer form a pure singlet.

THE XC FUNCTIONAL

$$E_{\mathrm{xc}}[n]$$

Exchange-correlation (XC) energy functional

A functional maps density to energy. Common approximations can miss static correlation; exact DFT need not.

H_2 exposes a central DFT challenge: the density can look right while the energy is wrong unless XC captures how electrons avoid each other.

Fixed nuclei; curves include nuclear repulsion. E_h: hartree, the atomic energy unit. Natural occupations: eigenvalues of the one-electron density matrix. Maps/formula: nonoverlapping 1s orbitals; singlet spin factor suppressed. Clouds are schematic.

Fuchs et al., JCP 122, 094116 (2005) · Burke: The ABC of DFT, §§4, 7, 11, 13 · Hehre et al., STO-3G (1969) · MIT: orbital symmetry · MIT: singlet spins